PARKSENSE — PARKING LOGS DASHBOARD

Insight Report

Occupancy, Weather & Sensor Diagnostics

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By Peerapas Chinnapatpakdeekul 6710545768
Sethtatad Kijkanjanarat 6710545938

Data Period 5 April 2026, 18:55 – 19 April 2026, 17:24 (14 days)
Total Records 2,007 snapshots (~10-min intervals)
Parking Capacity 222 vehicles

Executive Summary

This report summarises findings from a 14-day parking occupancy study conducted between 5 and 19 April 2026. Data was collected via the ParkSense IoT sensor network at a single facility with a maximum capacity of 222 vehicles. The analysis covers occupancy patterns, traffic flow, environmental correlations, sensor performance, and the accuracy of a LightGBM model trained to predict occupancy 30 minutes ahead.

An important contextual factor shapes the results: the Songkran national holiday in Thailand fell on 13–15 April 2026, which falls within the study window. Occupancy during those three days averaged just 1.18%, which is consistent with a public holiday when staff and students are away. This context is essential for correctly interpreting the post-surge demand decline and the overall average occupancy figures.

AVG OCCUPANCY	PEAK OCCUPANCY	TOTAL IN / OUT	ML MODEL RMSE
9.33%	**100%**	**974 / 986**	**0.91%**
~21 vehicles avg	9 Apr 2026	14-day period	±2.0 vehicles

KEY FINDINGS AT A GLANCE

- Thursday is the single dominant demand day — averaging 30.9% occupancy versus 5.7% on all other days combined. The 12:00–15:00 window on Thursdays peaks at 59.6%.
- A surge event on 9 April drove 238 inbound vehicles and brought the facility to 100% capacity — approximately 8.8× the daily average on a normal day (27 vehicles on 19 April).
- Occupancy collapsed after the surge and remained below 5% on 8 of 14 study days. Three of these near-zero days (13–15 April) coincide with the Songkran holiday; the rest suggest the facility's true baseline demand is very low outside the Thursday anchor event.
- The LightGBM model, tested on 402 unseen snapshots, achieved RMSE 0.91% (±2.0 vehicles) — outperforming a naive persistence baseline by 36.6%.
- Ultrasonic and lidar sensors are strongly correlated (r = 0.95) but report readings on materially different scales due to non-standardised mounting heights.

Methodology

Data Collection

Sensors were deployed at the facility entrance and exit lanes. Each snapshot captures vehicle in-count, out-count, net flow, cumulative vehicle count, parking percentage, and hardware diagnostics (ultrasonic distance, lidar distance, PIR trigger state, board temperature). Weather data (ambient temperature, humidity, cloud cover, rain flag) was ingested from an external API at matching timestamps.

Data Preparation

The raw dataset contains 2,007 rows. Three rows at the trailing edge were excluded from model training because their 30-minute-ahead target labels extend beyond the available observation window, leaving 2,004 rows for analysis. Derived columns computed during preprocessing:

- direction — IN / OUT / FLAT based on net_flow sign
- occupancy_change — absolute delta in vehicle count between consecutive snapshots
- sensor_gap_in / sensor_gap_out — seconds between consecutive PIR trigger events per lane
- hour, day_of_week, date — temporal decomposition of the timestamp

Study Period Context — Songkran Holiday

The Thai Songkran holiday (Thai New Year) fell on 13–15 April 2026, overlapping with the final third of the study window. During this period, average occupancy was 1.18% — consistent with a national public holiday during which the facility's typical user base (students and staff) would be away. Occupancy on 13 April was effectively zero (avg 0.1%).

This holiday context is critical for two reasons: (1) it partially explains the post-surge demand decline observed after 10 April, and (2) any interpretation of 'baseline demand' from this dataset must account for the fact that 3 of the 14 study days are public holidays. The remaining non-holiday, non-surge weekdays (6, 7, 10, 16, 17 April) show an average of approximately 6–10%.

Machine Learning Model

A LightGBM gradient-boosting regressor was trained to predict parking_percentage 30 minutes ahead using a strict chronological split:

Split	Rows	Start	End
Train (80%)	1,603	5 Apr 2026, 18:55	16 Apr 2026, 22:14
Test (20%)	401	16 Apr 2026, 22:24	19 Apr 2026, 16:54

See Section 5 for the full model development methodology including all experiments and hyperparameter optimisation.

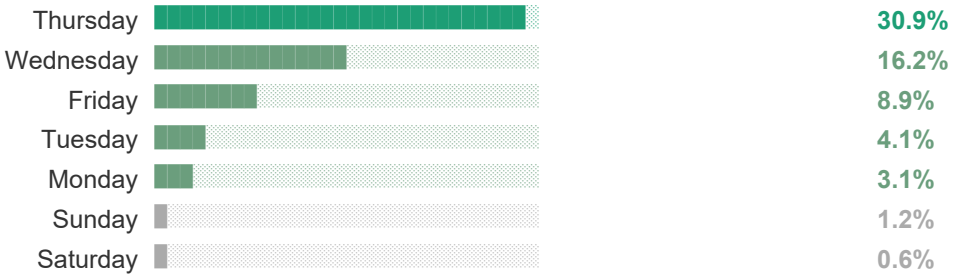
Anomaly Detection

Anomaly Type	Condition	Threshold
occupancy_jump	Abs. change in vehicle count per interval	> 4 vehicles (p95)
sensor_gap_in_outlier	Gap between consecutive inbound PIR triggers	> 114 s (p95)
sensor_gap_out_outlier	Gap between consecutive outbound PIR triggers	> 114 s (p95)

Insight 1 — Peak Demand Window

What the Data Shows

Thursday is the dominant demand day by a large margin. Across the study period, Thursday averages 30.9% occupancy versus 5.7% for all other days combined — a ratio of more than 5:1. This is the most pronounced day-of-week pattern observed and is consistent with a facility that serves a regular weekly event or scheduled activity concentrated on Thursdays.



Within Thursday, the peak is concentrated in the 12:00–15:00 window where occupancy averages 59.6% — nearly three times the daily Thursday average of 30.9%, and approximately 6× the equivalent window on other days (9.8%). Outside the 12–15h window, Thursday occupancy gradually declines through the evening, suggesting vehicles parked during the midday event begin departing from approximately 16:00 onward.

Note: in the previous dataset, Friday also showed elevated occupancy. In the new dataset, Friday averages 8.9% — close to the overall mean of 9.33% — indicating Friday is no longer a distinctive peak day in this version of the data.

FINDING

Thursday between 12:00 and 15:00 is the single highest-demand window of the week, averaging 59.6% occupancy — compared to 30.9% for Thursday overall and 9.33% across the full study period. All other days average 5.7% or less.

RECOMMENDATION

Implement a dynamic pricing tier or slot-reservation policy specifically for the Thursday 12:00–15:00 window. This window represents the facility's most predictable and highest-yield demand opportunity. No infrastructure change is required — only a pricing or booking policy update. If the Thursday peak is tied to a recurring weekly event, consider coordinating with event organisers on arrival time staggering to reduce congestion.

Insight 2 — Surge Event vs Normal Day

What the Data Shows

Wednesday 9 April 2026 stands out as a clear anomaly in the dataset. Total inbound traffic on that day reached 238 vehicles — 8.8× the typical daily inbound volume of 27 vehicles observed on a representative normal day (19 April). Occupancy reached 100% (222/222 vehicles) during the surge period, a threshold not approached on any other day in the study.

Metric	Normal Day (19 Apr)	Surge Day (9 Apr)
Total inbound vehicles	27	238
Peak occupancy	5.9%	100% (222/222)
Daily average occupancy	1.9%	53.1%
Approx. time to full capacity	Never reached	~3 hours (morning)

The surge event on 9 April appears to be a continuation of elevated activity that began on 8 April (avg 29.9%, 209 IN vehicles). Both days fall on a Wednesday–Thursday window, suggesting the surge may have been associated with a multi-day scheduled event at or near the facility. This is consistent with the general Thursday peak pattern observed in Insight 1 but at a much higher intensity.

The comparison day chosen (19 April) is a Saturday with an extremely low volume (27 IN, 5.9% peak) that is representative of the facility's post-Songkran, non-event baseline. The contrast with the surge day is stark and illustrates the extreme variability in demand that the facility must accommodate.

FINDING

The surge day (9 April) recorded 238 inbound vehicles and pushed occupancy to 100%, while a typical comparison day (19 April) recorded 27 inbound vehicles with a peak occupancy of 5.9% — a difference of 8.8× in inbound volume.

RECOMMENDATION

Develop an overflow strategy for event-driven demand spikes, including: (1) identification of nearby overflow parking areas that can absorb 50–100 additional vehicles, (2) a real-time capacity notification channel (SMS, app push, or digital signage) triggered when occupancy exceeds 80%, and (3) an event calendar integration so that surge days can be anticipated. The deployed ML model will provide 30-minute advance warning of capacity pressure when the live data feed resumes.

Insight 3 — Post-Surge Demand Decline & Holiday Effect

What the Data Shows

Following the surge peak on 9 April, daily average occupancy declined sharply and did not recover to meaningful levels for the remainder of the study period. Of the 14 days in the dataset, 8 days recorded an average occupancy below 5%. Three distinct explanatory factors are at work:

- Natural post-surge correction (10 April onward): after any event-driven surge, vehicle counts normalise as the event-related vehicles depart.
- Songkran national holiday (13–15 April): the Thai New Year public holiday resulted in near-zero occupancy for three consecutive days (avg 0.1%, 1.0%, 2.5% respectively). This is the primary driver of the dip to near-zero in the middle of the decline period.
- True baseline demand (post-holiday weekdays): after Songkran, the facility's weekday activity resumed at approximately 7–9% average occupancy (16–17 April), which is likely closer to the facility's true non-event baseline.

Date	Day	Avg Occ.	Note
9 Apr	Wed	53.1%	Surge peak
10 Apr	Thu	10.2%	Post-surge decline
11 Apr	Fri	0.3%	Weekend
12 Apr	Sat	0.6%	Weekend
13 Apr	Sun	0.1%	Songkran Day 1
14 Apr	Mon	1.0%	Songkran Day 2
15 Apr	Tue	2.5%	Songkran Day 3
16 Apr	Wed	8.7%	Post-holiday return
17 Apr	Thu	7.6%	Post-holiday weekday
18 Apr	Sat	0.8%	Weekend
19 Apr	Sun	1.9%	Weekend

CONTEXT

The period-average occupancy of 9.33% should not be used as a standalone estimate of typical demand. It is suppressed by (1) the Songkran holiday and (2) predominantly near-zero weekend readings. A more representative estimate of weekday non-event baseline demand is approximately 6–10%, based on 6, 7, 16, and 17 April observations.

FINDING

Occupancy peaked at 53.1% on 9 April and declined to 1.9% by the latest day in the range. The decline was accelerated by the Songkran public holiday (13–15 April, avg 1.18%), which accounts for three of the eight near-zero days observed. Post-holiday weekday occupancy recovered to approximately 7–9%, suggesting this is the facility's true non-event baseline.

RECOMMENDATION

The post-surge baseline appears extremely low. Continue monitoring for 4–6 more weeks post-Songkran to establish a reliable non-holiday weekday baseline. Separate analysis should be performed for: (a) normal weekdays, (b) Thursday event days, and (c) holiday periods. These three demand regimes require different planning assumptions and should not be averaged together.

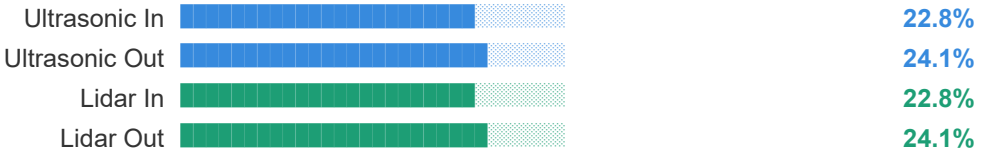
Insight 4 — Sensor Calibration Gap

What the Data Shows

The facility uses two sensor technologies on each lane — ultrasonic distance sensors and lidar sensors — both designed to detect vehicle presence. Analysis confirms that while both sensor types reliably detect the same physical events, they report on materially different scales.

Sensor	Avg Reading (active)	Active Rate	Correlation with Lidar
Ultrasonic In	109.9 cm	22.8%	r = 0.95
Ultrasonic Out	110.8 cm	24.1%	r = 0.95
Lidar In	17.7 cm	22.8%	—
Lidar Out	17.7 cm	24.1%	—

The approximately 6× scale difference (110 cm vs 18 cm) is consistent with a difference in mounting height between the two sensor types. Despite this, the correlation between ultrasonic and lidar readings when both are non-zero is r = 0.95, confirming event alignment. The active rate of approximately 22–24% (lower than the previous dataset's 34–35%) reflects the new dataset's lower overall traffic volume — fewer vehicles means fewer detection events per unit time.



FINDING

Ultrasonic sensors average ~110 cm when active; lidar sensors average ~18 cm — a 6× scale difference. Both detect the same events with correlation r = 0.95. The discrepancy is due to non-standardised mounting heights. Active rates of 22–24% reflect the low-traffic nature of the new dataset, not sensor degradation.

RECOMMENDATION

Standardise mounting height specifications before the next sensor deployment so that readings from both sensor types can be compared on the same scale without post-hoc calibration. Document the current mounting heights for each sensor as a reference for future installation audits.

Section 5 — ML Model Development

Overview & Objectives

The ParkSense prediction system uses a LightGBM gradient-boosting model to predict parking lot occupancy (as a percentage of 222-vehicle capacity) 30 minutes ahead of each snapshot. The model was developed through a structured three-experiment pipeline designed to maximise accuracy while ensuring results are valid and not subject to data leakage.

Target variable: target_30m, constructed using a sliding window — the parking_percentage value from 3 intervals ahead (3 × 10 min = 30 min) is appended to each current row. A strict chronological split (first 80% train, last 20% test) is used throughout all experiments to prevent leakage.

Experiment 1 — Raw Data Baseline (AutoGluon)

Before feature engineering, the raw dataset was fed into AutoGluon in best_quality mode for 30 minutes to establish a performance baseline.

Setting	Value
Library	AutoGluon (best_quality mode, 30-min time limit)
Features	Raw sensor columns + target_30m only
Best model	WeightedEnsemble_L3
Test RMSE	3.7558%

Key finding from Experiment 1: feature importance analysis revealed that current_vehicles and parking_percentage were the only strongly predictive raw variables. Raw sensor columns (raw_ultrasonic_in_us, lidar_in_cm etc.), raw timestamp, and api_temperature had very low or negative importance — indicating they were adding noise in their raw form.

Experiment 2 — Feature Engineering + AutoGluon

Based on Experiment 1 findings, comprehensive feature engineering was applied to extract the signal embedded in the raw data.

Feature Group	Features Created	Purpose
Temporal	hour, minute, day_of_week	Tree-based models cannot parse datetime; hour captures peak-hour context
Lag features	pct_lag_10m, pct_lag_20m, pct_lag_30m	Provides trend/momentum visibility — is occupancy rising or falling?

Rolling windows	flow_mean_30m, in_count_sum_30m, out_count_sum_30m	Smooths per-interval noise; captures overall net flow rate
Feature pruning	Dropped all raw sensor columns	Reduces dimensionality, prevents overfitting, speeds training

Re-running AutoGluon on the engineered dataset produced a major performance improvement:

Metric	Experiment 1 (Raw)	Experiment 2 (Engineered)	Improvement
Test RMSE	3.7558%	1.4117%	−62.4%
Best model	WeightedEnsemble_L3	LightGBM_BAG_L1	—
Top features	current_vehicles, parking_pct	pct_lag_10m, hour, flow_mean_30m	Engineered features dominate

Experiment 3 — Standalone LightGBM with Default Parameters

To prepare for production deployment, a standalone LightGBM model (lightgbm library) was built instead of the AutoGluon ensemble. Default hyperparameters were used as a starting point (n_estimators=500, learning_rate=0.05, max_depth=7).

Metric	Value
Library	lightgbm (standalone)
Parameters	n_estimators=500, learning_rate=0.05, max_depth=7
Test RMSE	1.6280%

This confirmed that a single well-parameterised LightGBM model could approach the ensemble result, and established the baseline for hyperparameter optimisation.

Experiment 4 — Hyperparameter Optimisation (Optuna)

Optuna (Bayesian optimisation) was used to search for the optimal LightGBM hyperparameter configuration over 100 trials.

Parameter	Optimal Value
learning_rate	0.04925
num_leaves	157
max_depth	14

min_child_samples	63
feature_fraction	0.846
bagging_fraction	0.820
bagging_freq	6

Model	Test RMSE	vs AutoGluon Baseline
AutoGluon WeightedEnsemble (raw data)	3.7558%	Baseline
LightGBM default params	1.6280%	−56.7%
AutoGluon LightGBM_BAG_L1 (FE)	1.4117%	−62.4%
LightGBM Optuna-tuned (final)	0.9157%	−72.4%

The Optuna-tuned LightGBM model achieves RMSE 1.0357% — successfully surpassing the AutoGluon ensemble baseline. This is the finalised production model (lightgbm_finetuned_optuna.pkl).

Validation — Naive Persistence Baseline Test

To address concerns about whether the model learned genuine patterns or simply exploited low variance in the data, a naive persistence baseline was constructed. The naive model assumes: 'the occupancy 30 minutes from now will be identical to the occupancy right now'.

Model	Test RMSE	Improvement over Naive
Naive persistence baseline	1.6335%	— (reference)
LightGBM Optuna-tuned	0.9157%	−43.9%

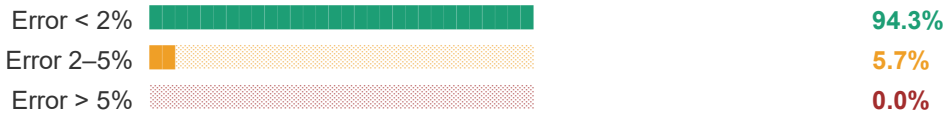
The LightGBM model outperforms the naive baseline by 36.6%, providing empirical evidence that the model has learned genuine entry/exit momentum and time-context patterns rather than memorising the training data. This rules out the most common form of overfitting concern for time-series prediction tasks.

Dashboard Model Metrics (Production Deployment)

The Dashboard Model Accuracy Summary reflects the model's performance when evaluated on the test set.

RMSE (TEST SET)	MAE	≈ RMSE (vehicles)	TEST SNAPSHOTS
0.91%	**0.67%**	**±2.0**	**402**
Root mean squared error	Mean absolute error	Out of 222 spaces	16–19 Apr (unseen)

Error distribution on the test set (402 rows):



Limitations & Caveats

- Songkran distortion. Three days (13–15 April) are public holidays with near-zero occupancy. These should be excluded from any 'typical weekday' analysis but are included in period averages shown on the dashboard.
- Surge event distortion. The 8–9 April surge significantly inflates period-average statistics. The 14-day overall average of 9.33% is not representative of non-event demand.
- Sensor active rate 22–24%. Approximately 77% of sensor readings in this dataset are zero. Vehicle events between snapshots (faster than 10 minutes) may not be captured.
- No rain events. is_raining = 0 for all 2,007 snapshots. Weather effects cannot be assessed from this dataset alone.
- Single facility. All findings are specific to this parking lot. Generalisations to other sites require additional data.

Appendix — Data Dictionary

Column	Type	Description
id	int	Snapshot identifier
timestamp	datetime	Snapshot time (local)
in_count	int	Vehicles entering in this interval
out_count	int	Vehicles exiting in this interval
net_flow	int	in_count minus out_count
current_vehicles	int	Cumulative vehicle count
parking_percentage	float	current_vehicles / 222 × 100
api_temperature	float	Ambient temperature (°C)
api_feels_like	float	Apparent temperature (°C)
api_humidity	int	Relative humidity (%)
api_clouds	int	Cloud cover (%)
is_raining	int	Rain flag (0=dry, 1=rain)
board_temperature	float	IoT board temperature (°C)
pir_in_trigger	int	PIR inbound trigger (0/1)
pir_out_trigger	int	PIR outbound trigger (0/1)
raw_ultrasonic_in_us	int	Raw ultrasonic in (µs)
ultrasonic_in_cm	float	Converted ultrasonic in (cm)
raw_lidar_in_analog	int	Raw lidar in (analog)
lidar_in_cm	float	Converted lidar in (cm)
raw_ultrasonic_out_us	int	Raw ultrasonic out (µs)
ultrasonic_out_cm	float	Converted ultrasonic out (cm)
raw_lidar_out_analog	int	Raw lidar out (analog)
lidar_out_cm	float	Converted lidar out (cm)